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Investigating the effect of mono di carboxylic acids as hydrogen bond donor on solvation of copper in choline chloride-based deep eutectic solvents

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ABSTRACT

Deep eutectic solvents (DESs) are green alternatives to highly acidic solvents or ionic liquids (ILs). The flexible complexation properties of eutectic solvents open their scope as a prospective dissolution medium for metal extraction. The present study investigates the range of monocarboxylic acids-choline chloride (Monocarboxy A-ChCl) and dicarboxylic acid-choline chloride (Dicarboxy A-ChCl) DESs as solvent media for copper oxide dissolution with a discussion on physical properties and their effect on dissolution. Thermal stability, freezing point, Kamlet Taft parameter, density, and viscosity of Carboxy A-ChCl DESs were experimentally determined.

The density of Carboxy A-ChCl DESs follows the trend: FA-ChCl < Acetic A-ChCl < OA-ChCl < MA-ChCl < CA-ChCl. The viscosity of Monocarboxy A-ChCl was observed to be lesser than Dicaboxy A-ChCl DESs. The polarity of Monocarboxy A-ChCl was observed to be in the range of 1.2–0.8, while Dicarboxy A-ChCl, polarity was greater than 1. Further, the effect of hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), KFT parameters, viscosity, fukui function of HBD, reaction time and temperature affecting the dissolution of Cu are elaborated in detail. Most dependant to least dependent parameters with respect to dissolution is recognized and some crucial conclusions were put together. Formic acid: Choline chloride (FA-ChCl) DES indicated the highest copper dissolution in the range $> 10^6$ ppm. Copper dissolution from ionic CuO to low polar Monocarboxy A-ChCl DESs was higher than in high polar Dicarboxy A-ChCl DESs. DFT calculation reveals that Formic Acid (FA) has the highest fukui functions compared to other carboxylic acids, which is directly reflected in the higher solubility of copper oxides. Additionally, the effect of reaction time and temperature on CuO dissolution in best-performing FA-ChCl DES was studied, 75 °C, 27 h, 100 °C, 18 h were chosen as the optimum time and temperature for maximum dissolution of Cu. The results cover important parameters of Carboxy A-ChCl DES, which could be helpful in metal extraction applications.

1. Introduction

Metal oxides exist naturally in the form of mineral ores such as hematite (Fe_2O_3), cuprite (Cu_2O), and pyrolusite (MnO_2). It is also in high composition in several waste materials such as lithium-ion batteries [1,2], spent catalysts [3–6], and as mining industry residues in the form of dust particles [7], slag [8], etc. Dissolution of metals has wide application during etching in the microelectronics industry, surface polishing of metals, synthesis of organometallic compounds, and extraction of metals from their ores [9]. In e-waste, commonly known as

Abbreviations: Acetic A, Acetic Acid; Acetic A-ChCl, Acetic Acid-Choline chloride; Adipic A, Adipic Acid; Adipic A-ChCl, Adipic Acid-Choline chloride; B3LYP, Beck's three parameter functional Lee Yang Parr; BA, Butyric Acid; BA-ChCl, Butyric Acid-Choline chloride; CA, Citric Acid; CA-ChCl, Citric Acid-Choline chloride; Carboxy A-ChCl, Carboxylic Acid-Choline chloride; ChCl, Choline Chloride; DENA, N, N-diethyl-4-nitroaniline; DESs, Deep Eutectic Solvents; DFT, Density Functional Theory; Dicarboxy A-ChCl, Dicarboxylic Acid-Choline chloride; DSC, Differential Scanning Calorimetery; EG, Ethylene Glycol; EG-ChCl, Ethylene Glycol-Choline chloride; FA, Formic Acid; FA-ChCl, Formic Acid-Choline chloride; FTIR, Fourier Transform Infrared Spectroscopy; HBA, Hydrogen Bond Acceptor; HBD, Hydrogen Bond Donor; ILs, Ionic Liquids; KFT, Kamlet Taft Parameters; LSER, Linear solvation energy relationship; MA, Malonic Acid; MA-ChCl, Malonic Acid-Choline chloride; Monocarboxy A-ChCl, Monocarboxylic Acid-Choline chloride; MPAES, Microwave Plasma atomic emission spectroscopy; NA, 4-nitroaniline; OA, Oxalic Acid; OA-ChCl, Oxalic Acid-Choline chloride; PA, Propionic Acid; PA-ChCl, Propionic Acid-Choline chloride; SA, Succinic Acid; SA-ChCl, Succinic Acid-Choline chloride; TGA, Thermo Gravimetric Analysis; U-ChCl, Urea-Choline chloride; UV-Vis, Ultraviolet-Visible; VFT, Vogler Fulcher Tamman.

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urban mines, metals exist in an elemental state, which essentially involves an additional oxidation step for further processing [10]. Recycling metals from these wastes is an urgent necessity to reduce the burden on mining industries and minimize the environmental effects.

Pyrometallurgy and hydrometallurgy are widely used technologies for the extraction of metals from ores and wastes such as spent catalysts and electronic wastes [11–13]. Pyrometallurgy is an intensive process that involves releasing harmful chemicals such as furan and dioxins [11]. Material loss occurs during the energy-intensive pyrometallurgical process [10]. Metals resale value and their concentration in ores or wastes play a crucial role in calculating the economic feasibility of the process. Since metal ores and urban mines are complex sources with low concentrations of metals, pyrometallurgy often rules out in terms of the economics and selectivity of metal extraction. Hydrometallurgy has been a preferred route for ores and other secondary sources with a low percentage of metals [12]. Hydrometallurgy involves the leaching of metals into a highly acidic or basic medium. However, hydrometallurgical processes discharge a colossal volume of aqueous waste into the environment during metal extraction, which should be further treated and purified before disposal [13,14] Another drawback is the occurrence of multi-metal pregnant solutions during the hydrometallurgical process due to the poor selectivity of metals. Additionally, individual pure metal is recovered using various other solvents and through a multi-step process.

In hydrometallurgy, metal extraction was conventionally done by halide leaching, acid leaching, cyanide leaching, etc., which affects the aquatic and soil ecosystem adversely [12–15]. Strong mineral acids such as HCl and H_2SO_4 were used in metal dissolution, with hydrogen evolution [16]. Currently, highly toxic aqua regia and cyanide solutions have been used at the industrial level, which is harmful to the environment [17]. Conventional mineral acids such as HCl, H_2SO_4, and HNO_3 are low-cost but hold a high environmental impact after usage. Economics vs environmental factors plays a major role in the development of any process; mineral acids, though, economical have been reported to be detrimental to the environment. ILs have high thermal stability, chemical stability, and tunability, making them an excellent alternative to existing conventional solvents. However, the high cost, the complexity of synthesis methods and the debatable toxicity of ILs make them a complex media for industrialization [18,19]. Greener alternatives to hydrometallurgy and ILs should be explored for the development of more sustainable processes.

DESs are promising alternatives to ILs and traditional solvents with easy preparation and no additional purification steps. Generally, DESs are prepared by mixing hydrogen bond donors (HBDs) with hydrogen bond acceptors (HBAs) at a particular ratio and a specific temperature. DES's physical properties and applications are reminiscent of ILs [20]. DESs functional groups such as carboxyl, and chlorides, which contains lone pair of electrons form metal complexes showing an efficient leaching medium [21]. DESs have been utilized in metal extraction from lithium batteries [22–25], copper ores [26], incinerated sludge [27], and NdFeB magnets [28]. The dissolution of elemental metals from waste to solution specifically in DESs is negligible without an oxidizing agent. Hence, the dissolution of metals involves an additional oxidation step with the aid of an oxidizing agent. Iodine in ethylene glycol: choline chloride DESs (EG-ChCl 2:1) has been used as a strong oxidizing power for leaching gold from gold ores [29]. Generally, organic acids are chosen as HBDs and choline chloride (ChCl) as HBA to synthesize acidic DESs. Acidic DESs, compared to mineral acids, have a moderate cost but are environmentally benign. Recovery of metals using mineral acids is significantly efficient, but the multistep precipitation and extraction are burdensome. DESs have low volatility hence can be integrated with precipitation or electrodeposition to recover metals and can be regenerated upto certain cycles [24]. Precipitation, electrodeposition and liquid–liquid extraction have been used extensively along with DESs leaching to recover metals from batteries and e-waste [24,25,30–32].

However, limited DESs as leaching agents are explored in the literature to extract metals from metal oxides or wastes and the impact of physical properties of DESs on its application. Therefore, exhaustive research is needed to explore the role of DESs physical properties, the effect of HBDs, and several other factors on the solubility of metals in the aqueous solutions, for efficient designing of the process. The dissolution of metal oxides in choline chloride (ChCl)-based DESs, prepared by varying HBDs such as ethylene glycol (EG), oxalic acid (OA), and malonic acid (MA), has been studied in the literature [33]. ChCl has been used chiefly as HBA because of its low cost, high biodegradability, and biocompatibility [34]. Commonly used HBDs are urea, carboxylic acids, alcohols, and amino acid [35]. It has been reported in the literature that Carboxy A-ChCl DESs owing to their acidic properties can dissolve more metal oxides when compared to their counterparts of amide and polyol based DESs [35]. Abbott et al. compared EG-ChCl with malonic acid: choline chloride (MA-ChCl) DES and observed high metal oxide solubility in MA-ChCl [36]. The high solubility of metal was correlated with the high proton activity of solvents which accepts oxide from metal oxide, hence, breaking the bond with metals forming a chlorinated metal complex [28]. Since acidic HBD in DESs has shown promising results for metal dissolution [36,37]. Hence, for this study Carboxy A-ChCl DESs are chosen for further evaluation. Pateli et al. discussed proton activity and complexing abilities of DESs on the solubility of d and p block metal oxides [38]. Metal solubility in low complexing EG-ChCl DESs is observed to be influenced by protonation or release of H^+ ions. The decrease in pH of the DESs resulted in the higher dissolution of metal oxide. In highly coordinating HBDs such as urea in urea-choline chloride (U-ChCl), surface complexation reactions have an important role in metal dissolution [38]. However, only a limited number of factors have been studied to understand the role of DESs on metal speciation.

Current research aims to compare different carboxylic acids as HBD, their effect on the physicochemical properties of Carboxy A-ChCl DESs, and in the solubility of copper oxide. The effect of HBDs, HBAs, polarity, reaction temperature and reaction time of DESs on the dissolution of copper oxide was studied in detail. Copper oxides with different lattice energies of Cu(I)O (1594 kJ mol^{-1}) and Cu (II)O (4050 kJ mol^{-1}) were selected for the study [38]. Copper is a primary constituent of e-waste, worth 10,960 million USD, therefore focus is on copper dissolution [39]. Additionally, CuO has a variety of industrial applications in catalysis, solar cells due to its unique properties and versatility as a material. CuO minerals such as cuprite, azurite, malachite are also important copper source [40]. Theoretical calculations were done using density functional theory (DFT) to optimize structures and to calculate the atomic charges and fukui function of Carboxy A-ChCl DESs. An intensive study on the properties of Carboxy A-ChCl DESs and their effect on solubility will help to better understand the green solvents and wisely utilize them in the development of metal extraction and separation processes.

2. Materials and methods

2.1. Chemicals and Standards

ChCl (98+%) was purchased from Alfa Aesar. FA, Acetic Acid (Acetic A), Propionic Acid (PA), Butyric Acid (BA), MA, OA, Adipic Acid (Adipic A), Succinic Acid (SA), copper(I) oxide, copper (II) oxide were obtained from Fisher Scientific. Double Distilled water with conductivity lower than 1 μS/cm was used for dilution. Reichardt dye, 4-nitroaniline, N, N-diethyl-4-nitroaniline of Alfa Aesar were used to calculate Kamlet-Taft parameters (K-T).

2.2. DESs preparation and analysis

Highly acidic DESs with pH in the range of acidic solutions were synthesized. The synthesis temperature of 50 °C was selected for Monocarboxy A-ChCl DESs. Dicarboxy A-ChCl based DESs were prepared at room temperature with 600 rpm stirring rate to avoid decarboxylation

reaction as reported by Delgado-Mellado and co-workers [41]. Different types of HBD were mixed with ChCl (HBA) in different molar ratios, as shown in Fig. 1, in sealed glass vials at a continuous stirring of 200 rpm at 50 °C to form eutectic mixture [36]. Judiciously selected HBDs and HBA with molar proportions are depicted in Table 1. Prepared DESs were stored at room temperature in a sealed container for further usage.

The density of DESs was calculated in Mettler Tolardo DE45 Delta Range Density meter. Change in viscosity of solvent was measured by Anton Paar MCR302 rotational and oscillatory rheometer. Thermal degradation of DESs was studied by Thermo Gravimetric Analyser in the range of 25 °C to 800 °C taking 10–20 mg of sample in a nitrogen atmosphere. DSC of samples was done in Perkin Elmer DSC 4000. Infrared spectra were recorded on a Nicolet iS50 FTIR spectrophotometer equipped with an ATR crystal detector. Absorption spectra of DES and metal concentrate were analysed by Agilent UV Cary 4000 UV–Vis. The water content of DESs was calculated using Karl Fischer Titration.

Table 1
Synthesized DESs.

Types of DES	Abbreviation	Ratio	DES
Monocarboxy A-ChCl DES	FA-ChCl	2:1	Formic Acid-Choline chloride
	Acetic A-ChCl		Acetic Acid-Choline chloride
	PA-ChCl		Propionic Acid-Choline chloride
	BA-ChCl		Butyric Acid-Choline chloride
Dicarboxy A -ChCl DES	OA-ChCl	1:1	Oxalic Acid-Choline chloride
	MA-ChCl		Malonic Acid-Choline chloride
	Adipic A-ChCl		Adipic Acid-Choline chloride
	SA-ChCl		Succinic Acid-Choline chloride
Tricarboxy A- ChCl DES	CA-ChCl	1:3	Citric Acid-Choline chloride

2.2.1. Solvatochromic probe behaviour of Carboxy A-ChCl DESs

Stock solutions of all probes were prepared by dissolving ethanol in amber glass vials and stored at 4 ± 1 °C. A calculated amount of dye was added to 1.5 mL quartz cuvette. Ethanol was evaporated using nitrogen gas. Further, a calculated amount of Carboxy A-ChCl was added to the cuvette and ultrasonicated for a few minutes for proper mixing. Reichardt dye, N, N-diethyl-4-nitroaniline (DENA), and 4-nitroaniline (NA) dyes were used in solvatochromic probes. Working concentrations of 1 mM for DENA and NA, and 5 mM for Reichardt dye were used to minimize the solute–solvent effect. Betaine 30 or Reichardt dye 30 are widely used as solvatochromic probes but it's zwitter ion nature is affected by the acidity of HBD [42]. DESs used in this research are highly acidic which might interfere with the nature of dye [42]. Therefore, Reichardt dye 30 has been used for this study [43]. Linear solvation energy relationship (LSER) is often denoted by the following equation:

$$v_{max} = v_{max0} + s(\pi^* + d\delta) + a\alpha + b\beta \quad (1)$$

$E_T(33)$ is evaluated by the lowest energy of charge transfer absorption [43] and is calculated by:

$$E_T(33)\ \text{kcal. mol}^{-1} = 28591.44/\lambda_{max} \quad (2)$$

$$\alpha = 0.0649\ E_T(33) - 2.03 - 0.71\ \pi^* \quad (3)$$

$$\beta = (1.035\ v_{N,N\text{-diethyl-4-nitoaniline}} + 2.64 - v_{4\text{-nitroaniline}})/2.80 \quad (4)$$

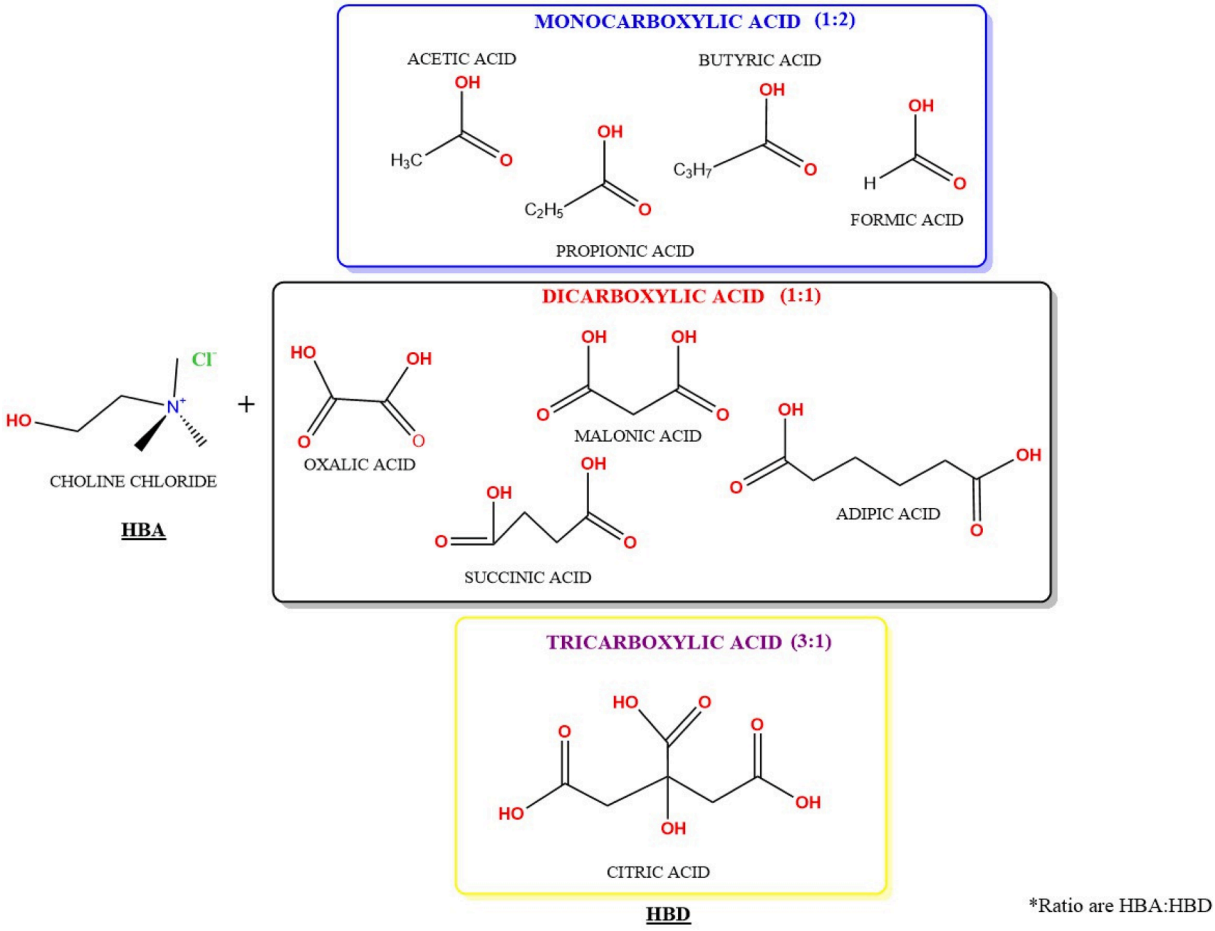


Fig. 1. Different Carboxy A-ChCl DESs used in this study.

$$\pi^* = 0.314\ (27.52\text{-}\nu_{N,N\text{-}diethyl\text{-}4\text{-}nitoaniline}) \quad (5)$$

$E_T(33)$ is in kcal mol^{-1}, v = 1000/λ_{max}, λ_{max} (nm) is the maximum wavelength for maximum absorption. Here, ν_{max} is the property of solute (solubility, reaction rate, etc) in each solvent, ν_{max0} is the same property in the reference state, π^* is dipolarity/polarizability, α represents hydrogen bond donating ability and β represents hydrogen bond accepting ability, δ is correction factor [44]. *a, b, d, s* are the independent solvent coefficients.

2.2.2. Computational details

Comprehensive and systematic analysis of these DESs was performed by DFT using Dmol^3 module by Beck's three-parameter exchange functional and Lee Yang Parr correlation parameter [45]. B3LYP method was used to investigate the structures and geometrically optimize Carboxy A-ChCl DESs [46]. Density functional B3LYP is a successful method to predict accurate molecular structure and vibrational spectra in dipole–dipole forces and hydrogen bonding with lowered error [47]. To ascertain whether elements participate in organic systems as electron donors (loss) or receivers (gain), DFT simulations using fukui functions (eq. (6)–(8) have been used [48]:

$$\text{Nucleophilic attack } f_A^{+} = q_A(N+1) - q_A(N) \quad (6)$$

$$\text{Electrophilic attack } f_A^{-} = q_A(N) - q_A(N-1) \quad (7)$$

$$\text{Radical attack } f_A^{0} = (q_A(N+1) - q_A(N-1))/2 \quad (8)$$

Here, f_A signifies a change in the electron density of atom A when the number of electrons changes, q_A represents the charge on atom A, and *(N-1), N, (N + 1)* signifies anionic, neutral, and cationic species, respectively.

2.3. Solubility of copper oxide

The solubility of copper oxides in different DESs was experimentally determined. 0.2 g metal oxide was mixed with 5 mL of DES at 50 °C and 500 rpm for 24 h [36]. The solution was first centrifuged and then filtered with a syringe filter of pore size 0.2 μm. The metal concentration of the filtered sample was determined by microwave plasma atomic emission spectroscopy (MPAES) diluted with doubly distilled water. Standards of copper used are Mixture 2 of Agilent Technologies. The samples were diluted up to 1000 times to have a final metal concentration lower than 50 ppm. Triplicate experiments were conducted, and averaging was done to reach a proper conclusion.

3. Results and discussions

3.1. DES formation

Monocarboxylic Acids (FA, Acetic A, PA, BA) were mixed with ChCl in 2:1 M ratio and remained liquid at room temperature. In the case of Dicarboxy A- ChCl, MA-ChCl and OA-ChCl were liquid at room temperature, while Adipic A-ChCl and SA-ChCl solidified. Dicarboxylic Acids (OA, MA, Adipic A, SA) were mixed with ChCl in 1:1 ratio. Citric acid (CA) was mixed with ChCl in 1:3 ratio.

Structural inspection of Monocarboxy A/Dicarboxy A-ChCl based DESs was analyzed using IR spectra. FTIR spectra of synthesized DES were evaluated to highlight hydrogen bonding in the eutectic systems. It was observed that the nature of HBDs and molar ratios has contributed to stretched vibrations because of a change in molecular bonding. Shifting of IR spectra of DES with respect to initial components signifies a substantial hydrogen bonding of ChCl and Monocarboxy A/Dicarboxy A. Fig. S1 (a-d) illustrates FTIR spectra of Monocarboxy A-ChCl, Dicarboxy A-ChCl DESs.

Choline chloride have a wide band from 3621 cm^{-1} to 2980 cm^{-1} portraying presence of OH group in the vibration stretching [49]. Quaternary ammonium ion of choline chloride is represented around ~ 928–980 cm^{-1}. Vibrations for Monocarboxy A account for OH stretching, C—H stretching, sp^2 C—O stretching, and sp^3 C—O stretching, respectively [46,50]. Shifting of OH vibration from 3027 cm^{-1} in ChCl to 2920 cm^{-1} in FA-ChCl indicates strong hydrogen bonding. sp^2 CO vibration shift was also observed in DESs, and C—H vibration shifted from 3027 to 2962 cm^{-1}. Similarly, if we consider Acetic A-ChCl, a positive shift in frequency for the C—H vibration in ChCl compared to the vibration in Acetic A-ChCl indicates the involvement of hydrogens of the methyl group in hydrogen bond formation. Shifting of the OH vibration from 3027 cm^{-1} in ChCl to 2920 cm^{-1} in DES indicated the formation of new hydrogen bonds. As the methyl group in HBDs increases, a greater shift in OH stretching was observed, which signifies the formation of hydrogen bonds with an increase in the H atom of a methyl group. If OH stretching of OA-ChCl and FA-ChCl are compared, there are no significant shifts in OH stretch in both acids, which signifies that the addition of COOH does not contribute much to the hydrogen bond.

Dicarboxy A-ChCl DESs have a broad band in the region of 3860 to 2270 cm^{-1} due to OH stretching and at 1718 cm^{-1} due to C = O stretching vibrations. C-N stretching peak of ChCl vanished in OA-ChCl, which signifies the movement of C = O under the effect of strong electronegative Cl$^-$ ion. OH peak of OA-ChCl at 3392 cm^{-1} indicates the presence of intramolecular hydrogen bonding between ChCl and OA.

3.2. Physiochemical property of DESs

3.2.1. Freezing point and thermal stability

The freezing point of a eutectic mixture is a crucial parameter to estimate the magnitude of the interaction between two molecules in a binary mixture. Generally, the dip in freezing point is directly proportional to the strength of the interaction - the greater the interaction, the greater the drop in freezing point [20,35]. According to Abbott et al., for OA-ChCl eutectic mixture, the individual melting point of OA and ChCl is 101.5 °C and 302 °C, respectively [21]. However, when ChCl is mixed with OA acid at 1:1 M ratio, the freezing point of DESs was observed to be in the range of 34 °C – 30 °C in different studies [37]. The molecular weight of carboxylic acid has a significant effect on the depression of freezing point [51]. If the freezing point of an individual component is considered, much variation was observed in the literature, ChCl freezing point was in the range of 216.70 °C −302 °C [41]. Reviewing HBD freezing point, Monocarboxy A-ChCl solidifies at a lower temperature than Dicarboxy A-ChCl. DES freezing points are generally less than individual components, depending on the mole ratio (HBD:HBA) and molecular structures [39]. DESs are especially desirable when their freezing points are less than 50 °C as they will have a greater liquid range and could be used as liquid catalysts or solvents in a variety of applications [33]. The freezing point of DESs was measured using Perkin Elmer DSC 4000. Initially, 10–15 mg were placed in a nitrogen atmosphere for all samples. Heating from −20 °C to 80 °C at the rate of 5 °C/min, equilibrating for 3 min and cooling from 80 °C to −20 °C at the rate of 1 °C/min. Table S1 illustrates the freezing point of HBD, from the above-mentioned program of DSC freezing point of PA and BA- ChCl DES cannot be determined. It was assumed to be lower than –20 °C. FA-ChCl and Acetic A-ChCl freezing points were –18.2 °C and –10.6 °C, respectively. The freezing point of Monocarboxy A-ChCl DESs has not been mentioned in the literature. The dip in freezing point of Monocarboxy A-ChCl DES is relatively less than Dicarboxy A-ChCl based DES. Hence, the hydrogen bond strength of Monocarboxy A-ChCl are relatively less as compared to Dicarboxy A-ChCl based DESs [35]. Its low freezing point is cutting edge for its wider applications in metal extraction.

The thermal stability of Carboxy A-ChCl DES was determined using Thermo Gravimetric Analysis. It is an important factor in the estimation of the maximum operating temperature of the solvent for its application. The thermal stability of Monocarboxy A-ChCl DES has a very minimal discussion in the literature. Fig. S2 (a) and (b) depict two weight loss

stages for Monocarboxy A-ChCl and Dicarboxy A-ChCl DESs, respectively. The degradation temperature of SA-ChCl and Adipic A-ChCl DES was observed to be near 180 °C and 190 °C, respectively. Monocarboxy A-ChCl DES has thermal stability upto 115 °C with an average weight loss of 5–10 %. Maximum decomposition occurs in the range of 115–220 °C. It has been observed from TGA of Monocarboxy A-ChCl at a temperature less than 100 °C, the average weight loss was 5 ~ 6 % which correlates with the water content of DESs. The water content of DES is reported in Table S2. Considering the freezing point and degradation temperature of Monocarboxy A-ChCl DESs, it is suitable for temperatures of less than 0 °C to 100 °C applications. Moving to Dicarboxy A-ChCl DESs, heating is required to keep the solvent in a liquid state at room temperature except for MA-ChCl, which remains in a liquid state until 18 °C. The weight loss at temperatures below 110 °C was due to moisture removal. Isotherm analysis of all DES was done at 25 °C, 60 °C, 80 °C and 100 °C for 2 h. Average weight loss was 2 % at 25 °C; 2–4 % at 60 °C and 80 °C. However, at 100 °C, an average weight loss of ~ 10 % was observed in all DESs. The study on freezing temperature and decomposition temperatures could help to identify the operational temperature range of the Carboxy A-ChCl DESs. It could be utilized for scale-up reactor designing of the process.

3.2.2. Kamlet Taft parameter

Hydrogen bond acidity (α), hydrogen basicity (β), and polarity (π) are some of the important parameters to be considered in understanding solvent properties and interactions, assisting in the prediction of DESs performance in its applications. Although, studies on KFT parameters of Carboxy A-ChCl DESs are limited [42–44,52]. Polarity measures the interactions, and ability of the solvent to alleviate charge via the dielectric effect [43]. The α depends on the cation and β on the anion part of DESs [53]. Dwamena et al. reported that polarity and acidity are independent of temperature [43]. Yet, Pandey et al. concluded that temperature does not change the polarity and basicity of DESs [42]. Florindo et al. evaluated intermolecular interactions between HBD and HBA by calculating the Kamlet Taft parameter of choline cation and carboxylic acids as anions, mainly MA-ChCl [54]. It was observed that the chemical structure of HBA and HBD have a significant effect on solvatochromic probe behaviour. Fig. 2 illustrates the calculated Kamlet Taft parameters of the synthesized DESs. Dipolarity/ polarizability of solvent (π) and hydrogen bond acidity (α) of Carboxy A-ChCl DESs decreases with an increase in alkyl chain length gesturing the prominent role of HBDs in assessing KFT parameters. Table S3 reflects the value of $E_T(33)$, π, β, α of Carboxy A-ChCl DESs. Dicarboxy A-ChCl DESs were observed to have higher α than Monocarboxy A-ChCl DESs due to the presence of two COOH groups in HBD. The high acidity of DESs owes to the high acidity of organic acids which are used as HBDs. E.g., OA has 1.3 pH as compared to FA which has 2.38 pH in 100 mM solution. Similarly, E_T (33) value of OA-ChCl was observed to be 69.733 kcal.mol^{-1} while FA-ChCl has 64.979 kcal.mol^{-1}. $E_T(33)$ is directly related to polarity, which signifies from the above result that the addition of COOH group increases the polarity of the solvent [55].

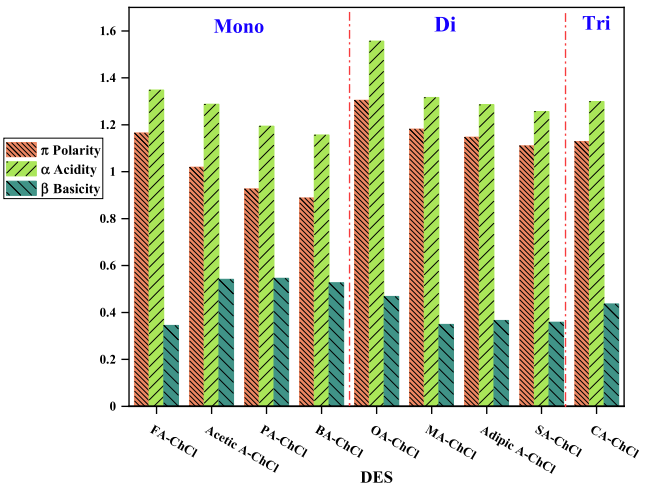


Fig. 2. Kamlet-Taft parameters of Carboxy A-ChCl DESs (Error = ±3.14%).

3.2.3. Density

The density of a solvent is an important parameter for process design, equipment size, and cost estimation [52]. Equation of state could be estimated by studying the effect of pressure and [56]. Table S4 highlights the density of Carboxy A-ChCl DES with the change in temperature, HBD, and mole ratio. A combination of HBA and HBD alters the density of DES. It was observed that DES with $ZnCl_2$ as HBA in combination with any HBD have densities in the range of 1.3–1.6 g/cm^3 [57]. Basaiahgari et al. compared the density of DES composed of alcohols (EG, glycerol, triethylene glycol) and choline chloride [58]. The density of EG-ChCl was examined to be lower than Glycerol-ChCl. The author concluded that the addition of OH group results in the formation of hydrogen bonds, which decreases the free volume [58].

The density of DESs depends on the temperature, and it relies on the presence of vacancies in the DESs network [35]. Change in density with change in HBD and molar ratio of HBD has been reported by several researchers [41,59]. Density vs. temperature profiles of Monocarboxy A, Dicarboxy A, and Tricarboxy A DESs are depicted in Fig. S3. CA-ChCl-based DESs exhibit higher density than water at room temperature and are mostly higher than ILs and several other organic solvents [21]. Thermal expansion coefficient (α), molecular volume (V_m), standard molar entropy (S^0), and lattice energy (U_{POT}) can be evaluated using the density of DES by below equations:

$$\ln\rho = b - \alpha T \tag{9}$$

$$V_m = \frac{M}{\rho} \tag{10}$$

$$S^{\circ} = 1246.5.V_m + 29.5 \tag{11}$$

$$U_{POT} = 1981.2\left(\frac{\rho}{M}\right)^{1/3} \tag{12}$$

Here, b represents the constant, α thermal coefficient, T temperature, and M is the average molecule weight of the component.

The thermal coefficient (α) of Carboxy A-ChCl DES, which is dependent on temperature, could be an important parameter to understand DESs dynamics and viscosity. The α of ILs also correlates with its ability to dissolve CO_2 by Kilaru and Scovazzo [60]. The α value is calculated by equation (9) which ranges from 1×10^{-4} to 10×10^{-4} K^{-1} for Carboxy A-ChCl as shown in Table S4. The thermal coefficient of FA-ChCl and Acetic A-ChCl is low compared to OA-ChCl, MA-ChCl, and CA-ChCl. It follows the trend FA-ChCl < Acetic A-ChCl < OA-ChCl < MA-ChCl < CA-ChCl. The CA-ChCl has more free volume and thus controlling the temperature of CA-ChCl DES, will lead to more expansion and more accommodation of molecules in between DES structures at high temperatures. Basaiahgari et al. observed the increase in expansion with an increase in temperature caused by higher kinetic energy molecules [58]. The higher expansion indicates the higher free volume in between molecules [58].

V_m signifies the volume occupied by each mole of DES. It is calculated by equation (10). Calculated values of V_m of Carboxy A-ChCl DESs are listed in Table S5. The increase in V_m was observed with an increase in the molecular size of HBD/HBA FA-ChCl < Acetic A-ChCl < PA-ChCl < BA-ChCl. The addition of –COOH in OA-ChCl shows an increase in V_m as compared to FA-ChCl. Hence, the densities of Dicarboxy A–ChCl DESs are higher than Monocarboxy A-ChCl DESs.

3.2.4. Viscosity

Flow characteristics of DES can be estimated by understanding the viscosity of DES which indirectly affects the mass transfer in separation [52,61]. The viscosity of Carboxy A-ChCl DESs was measured using Anton Par M302 rheometer using cone plate DIN 53019 rheometer geometry. At a shear rate of 1000 1/s varying temperature from 20 to 60 °C. Vogler Fulcher Tamman (VFT) equation and Arrhenius equation are mostly applied to DESs [35].

The viscosity of DESs with mono, di, and tri-carboxylic acids as HBDs was measured as shown in Table S6. It was observed that mixtures are exponentially dependent on the temperature following the Arrhenius equation:

$$\ln\eta = \ln\eta^{\circ} + \frac{E_{\eta}}{RT} \tag{13}$$

Profiles of viscosity/T of the Carboxy A-ChCl DESs have been reported in the following section in the temperature range of 303–333 °C. The viscosity of almost all Carboxy A-ChCl DESs was high (>100 cP) at room temperature. Larger viscosity is a result of the large size of solvent ions and stringent hydrogen bonding network. The intense hydrogen bonding results in low molar volume and low free volume in liquid.

FA-ChCl has the lowest activation energy compared to a series of Carboxy A-ChCl DESs thereby, making it most suitable for scale-up extraction processes. Dissolution of copper in lower viscous solvent FA-ChCl was maximum because of lower mass transfer resistance, while it decreases on moving to higher viscous DESs. High viscosity of the solvent is attributed to the low dissolution rate of metals in the solvent [9]. However, the addition of water in DES weakens hydrogen bonding, similar effect happens when increasing temperature where the vibrational energy of the component increases and weakens the hydrogen network [62]. Ma et al. confirm that a moderate amount of water establishes hydrogen bonds, and promotes delocalization due to the release of Ch^{+} and oxalate enhancing surface area for dissolution [63].

3.3. Metal dissolution in DES

3.3.1. Effect of HBDs

HBDs are classified into various groups according to the functional group, mainly alcohols, organic acids, and sugars [35]. Mainly, ChCl was used as HBA because of its low toxicity, and high biodegradability [36,56]. Abbott et al. tested U-ChCl, EG-ChCl, and MA-ChCl based DES for metal dissolution [36]. It was concluded that MA-ChCl exhibits the highest solubility of copper oxide compared to U-ChCl DES [36]. Multiple publications have reported the advantages of tuneable solvent properties of DESs by changing HBDs [21,36,37]. Present study evaluates the behavioural change in solubility of CuO in the range of carboxylic acids (FA, Acetic A, BA, PA; MA, OA, Adipic A, SA)-ChCl DESs. It was observed that in FA-ChCl maximum copper solubility occurs in Cu (I) and Cu (II) states. FA-ChCl, Acetic A-ChCl have much higher solubility (>10^6 ppm) of copper as compared to OA-ChCl, and MA-ChCl DESs (~10^4 ppm). The solubility of copper from its salts decreases as the alkyl chain length of HBD increases. The intense hydrogen bonding with an increase in alkyl change length in HBDs hinders the formation of metal complexes with the carboxyl group of DESs, resulting in lower solubility [64]. Cu (II)O and Cu (I)O dissolution in Carboxy A-ChCl DESs is illustrated in Fig. 3. In Monocarboxy A-ChCl DESs, the solubility of copper in Cu (II) state is higher than Cu(I) state. However, there is a slight difference in the solubility of copper in both states in almost all DES due to slight variations in Gibbs free energy, and lattice energy of both salts [38]. Changes in metal speciation and coordination number of metal ions are additional factors that influence the solubility of metal oxides, which is assumed to be dominant in FA-ChCl, BA-ChCl, and OA-ChCl DESs. Acetic A- ChCl DES as well as Dicarboxy A -ChCl showed higher solubility from Cu (II)O salt, which shows the dominance of the effect of lattice energy on the solubility. The lattice energy of Cu (I)O is lower as compared to Cu (II)O. Hence, Cu (II)O is the stable form of copper with more negative Gibbs free energy as compared to Cu (I)O [38]. When HBD in DES is changed from OA to MA, the solubility of copper from Cu (II)O decreases. In Monocarboxy A: ChCl DESs with the addition of CH_2 ion, there is a rapid decrease in the solubility of copper.

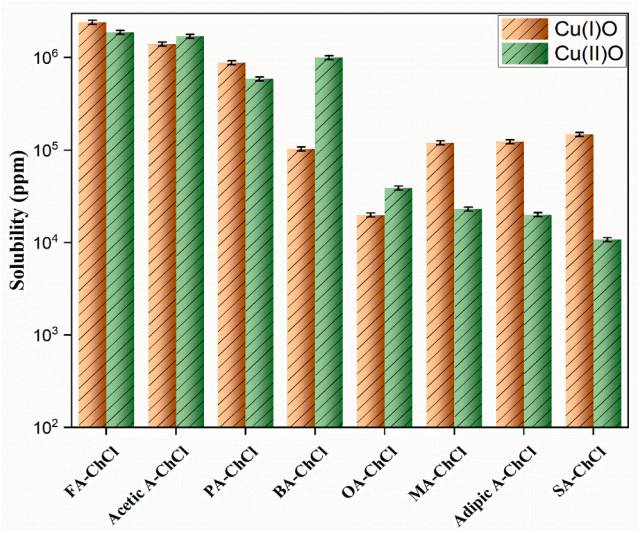


Fig. 3. Solubility of copper oxide in different Monocarboxy A-ChCl and Dicarboxy A-ChCl DES at 100 °C, 21 h; S:L ratio = 0.2 g/10 mL.

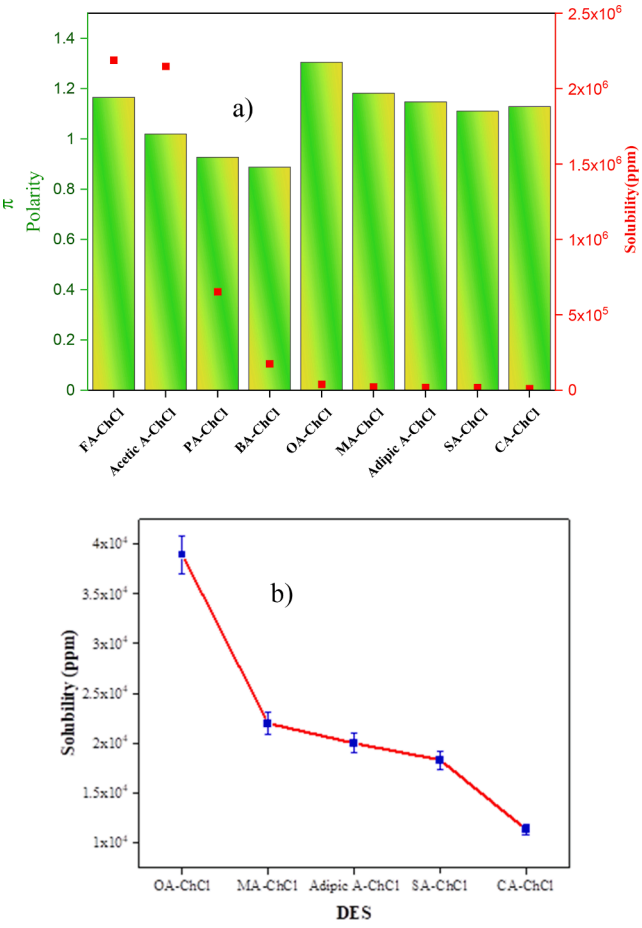


Fig. 4. (a). Relation between polarity and solubility of copper from Cu (II)O in Carboxy A-ChCl DESs (b). Solubility of Dicarboxy A-ChCl DESs.

3.3.2. Effect of polarity

The dependency of copper oxide solubility on the polarity of DESs is related in Fig. 4. However, no such relation was observed between the solubility and polarity of DESs. Interestingly, in each class of DESs the solubility decreases with the addition of CH_2 e.g., in Monocarboxy A-ChCl DESs, the solubility of copper oxide decreases as HBDs change from FA to BA. More polar FA-ChCl shows higher solubility than less polar BA-ChCl. However, shifting from Monocarboxy A to Dicarboxy A ChCl DESs, an increase in polarity was observed. Nevertheless, copper oxide solubility in OA-ChCl is less than FA-ChCl. The addition of –COOH group in Dicarboxy A-ChCl, led to a more stringent hydrogen bonding network and indirectly hinders the metal bond formation ability with DESs.

3.3.3. Effect of HBA

Complexing ligands have a significant effect on improving dissolution in DESs along with pH and other factors [38]. The addition of HBA in DESs plays a significant role. Fig. 5 suggests that the solubility of copper oxide in 1 N organic acid is low as compared to the respective Carboxy A-ChCl DES. Chloride ion in ChCl plays an important role in forming metal complexation [38]. Acetic A forms a precipitate with copper oxide and results in cupric acetate. In Acetic A-ChCl, copper dissolute in the solution due to the presence of chloride ion of ChCl. This confirms that HBA enhances the dissolution of metal in solution either by charge delocalisation or its tendency to form metal chlorate complex. For Cu^{2+}, Cl^- is an important ligand which is the reason for increase in dissolution on addition of HBA having chloride ion. Cu (II)O manifests higher dissolution in FA-ChCl and FA as compared to Cu(I)O. In case of Acetic A-ChCl, Cu(I)O precipitated while in its DES, metal remained in liquid state. However, Cu (II)O solubility slightly decreased in Acetic A-ChCl DES. Table S7 illustrates copper solubility in Dicarboxy A and there DESs.

3.3.4. Fukui function of HBD (using DFT calculation)

Identifying electrophilic and nucleophilic region of HBD via DFT could be an efficient tool to screen DESs for the dissolution of metal [65]. Fukui function is an important descriptor to identify nucleophilic, electrophilic, and radical attack region, by calculating change in electron density caused by change in number of electrons. It was observed that stronger the reactivity, larger the value of fukui function f_A^+/f_A^-. Theoretical calculation of Hirshfield charge distribution on HBD by natural population analysis (NPA) were used to calculate fukui functions thereby, identifying active sites. Atomic charges derived from Hirshfield

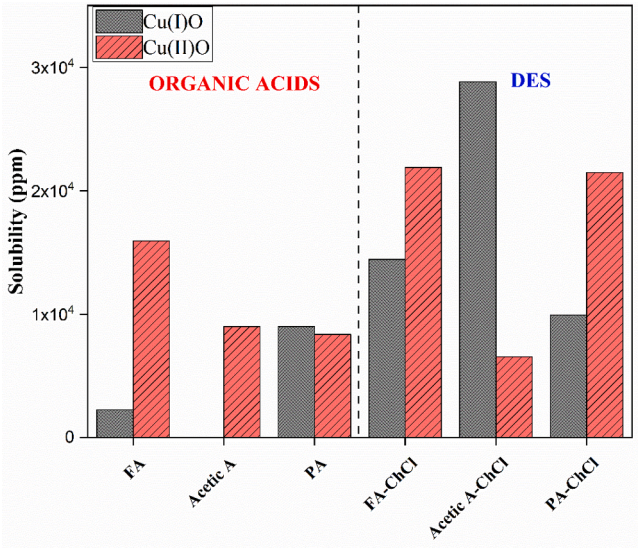


Fig. 5. Solubility of Cu(I)O and Cu (II)O (±3.97%);0.2 g /5mL in 1 N carboxylic acid solution and in DES at 100 °C and 24 h.

are more reliable and realistic as compared to mulliken charge to calculate condensed fukui function [66]. Fukui calculation of carboxylic acid shown in Fig. 6 has been done and tabulated in Table S8.

Carboxylic acid has C═O and one O—H bond. The C═O bond is polarised, as oxygen has a higher electronegativity than carbon. Thus, carbonyl carbon act as an electrophilic and carbonyl oxygen as nucleophilic [67]. The charge on carbonyl group oxygen was calculated using Fukui functions by DFT according to equation (6)–(8). First, the carboxylic acids are geometrically optimized as shown in Fig. 6 and further Fukui calculation was done to determine the f_A^+, f_A^- *and* f_A^0 of DES. The calculated order of f_A^+ in Monocarboxy A is in the trend: O atom of FA > O atom of Acetic A > O atom of PA > O atom of BA. This also goes with the dissolution of copper which decreases as we go through the series. It was also observed that (f_A^+) of O (3) of FA > O (3) of Acetic A > O(3) of OA > O(3) of MA. To accelerate metal dissolution, time and temperature could be modified for better and efficient extraction [38], which is illustrated in further sections. Table 2 highlights the calculated nucleophilic attack (f_A^+) value of carbonyl oxygen of carboxylic acids. Fukui calculation showing nucleophilic, electrophilic, and radical values are shown in Table S8.

3.3.5. Effect of time and temperature

After careful evaluation of all physical parameters in the above sections, FA-ChCl was observed to have the lowest viscosity and highest reducing power with maximum copper dissolution. In this section, CuO is dissolved in FA-ChCl DES at different temperatures and time to evaluate the optimum operating condition for the dissolution reaction as shown in Fig. 7. Copper oxide is chosen as metal oxide for this study as copper in e-waste is in CuO form after oxidation. At 100 °C and 125 °C, the dissolution of copper from copper oxide is almost similar for time greater than 18 h. It was also observed that at 50 °C, 100 °C and 125 °C equilibrium solubility has been achieved in 18 h. However, for 25 °C, 15 h was the optimum time whereas for 75 °C the equilibrium is reached in 27 h. After 24 h, for temperatures above 50 °C overall solubility of copper was similar irrespective of temperature. Based on thermal stability and efficient solubility 75 °C, 27 h, and 100 °C, 18 h is chosen as an optimum time and temperature for maximum copper dissolution.

3.4. Mechanism

The mechanism of dissolution of metals is often discussed in the literature. Abbott et al. stated that metal chlorate species are formed $[M_xClO(Carboxylic\ acid)]^-$ in the reaction of metal with DES [21]. Carbonyl oxygen has a lone pair of electrons, it forms a coordinating bond with metal as (M—O or M—OH). However, Shishov et al. revealed carboxylic acids as the chelating agent, promoting mass transfer from metal oxides to DESs [68]. Chelate complex formation of Carboxy A-ChCl DES shows more affinity towards metal dissolution. Organic acids like OA, CA and lactic acid have been promising lixiviants for metal oxides because of the formation of complex with metals [68]. Complexation of organic anions requires two stages where first carboxyl acid is chemisorbed in the active sites of metal oxide. If the metal ligand-carboxyl complex is more stable than metal-OH, the complex anion is substituted by OH sites and the reaction is pushed forward [69]. The surface of metal oxide behaved as an electron pair donor; the system interface subsequently emerged as a positively charged layer.

$$RCOOH \leftrightarrow H^+ + RCOO^- \textit{ Ionisation of acid} \quad (14)$$

$$Cu—O + H^+ \leftrightarrow Cu\text{-}O—H^+ \textit{ Protonation of oxygen} \quad (15)$$

Metal oxide bond is weakened by the protonation process and hence results in the dissolution of metal in solution. Now, the metal surface hydroxyl group becomes active sites for organic.

$$Cu—O —H^+ + RCOO^- \leftrightarrow [Cu\text{-}RCOO] + H_2O \textit{ Surface Complexation} \quad (16)$$

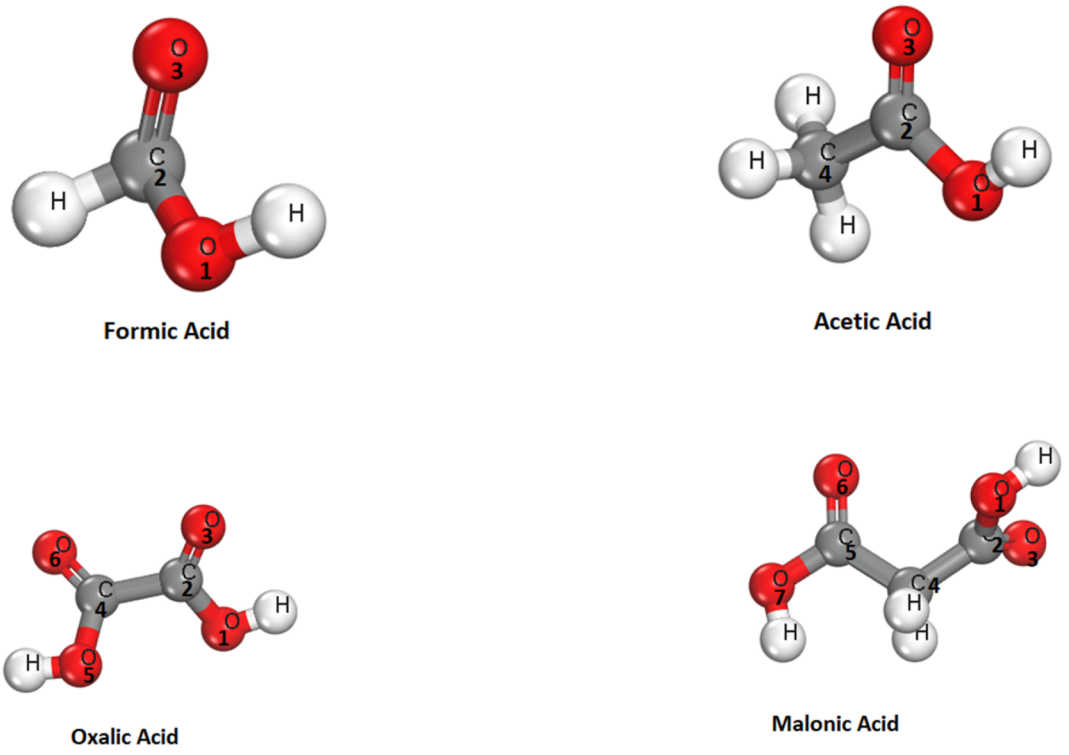


Fig. 6. Geometrically optimized structure of carboxylic acids.

Table 2
Calculated Nucleophilic attack (f_A^+) value of carbonyl oxygen atom of carboxylic acids.

Acid	Atom	Nucleophilic attack (f_A^+)
FA	O(3)	0.259
Acetic A	O(3)	0.224
OA	O(3)	0.183
	O(6)	0.183
MA	O(3)	0.087
	O(6)	0.047

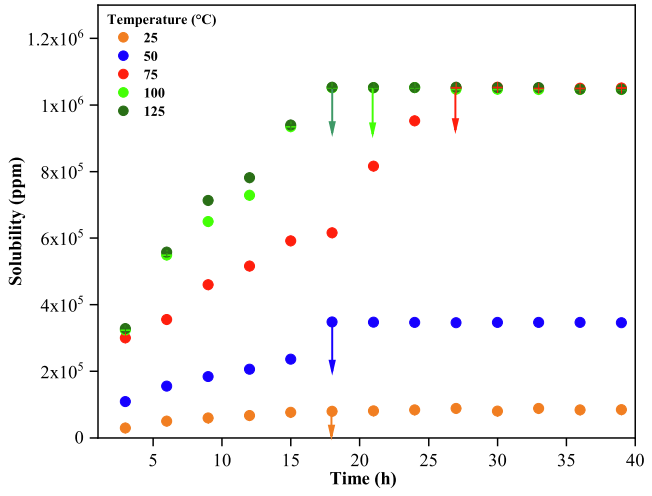


Fig. 7. Effect of time and temperature on dissolution of copper from CuO in FA-ChCl (S:L ratio = 0.2 g/10 mL; Solubility error ± 2.65%).

FTIR spectra of neat FA-ChCl show a broad peak at 3367 cm $^{-1}$ and Acetic A-ChCl has a slight stretch and bend at 3158.28 cm $^{-1}$, respectively in Fig. 8. After copper dissolution, there is a shift in wavelength to 3341.5 cm^{-1} in FA-ChCl and 3167 cm^{-1} in Acetic A-ChCl. It indicates an increment of molecular association between —OH group and metal ion. Further, —COO$^-$ stretching at 1715 cm^{-1} in neat FA-ChCl and shifting to 1723 cm^{-1} indicates the coordination of metal ions with the carboxyl group. However, in Acetic A-ChCl it was not visible in FTIR spectra.

4. Conclusion

The physiochemical properties of mono/di Carboxy A-ChCl based DESs have been experimentally determined. Particularly, freezing point, thermal stability, KFT parameters of DESs were determined along with the change in density and viscosity of DESs with respect to temperatures. Monocarboxy A-ChCl based DESs were stable at room temperature, while the Dicarboxy A-ChCl remained in solid state. Density and viscosity of Dicarboxy A-ChCl was higher than Monocarboxy A-ChCl and is dependent on temperature. Among all considered DESs, FA-ChCl have lowest viscosity and density as compared to other Carboxy A-ChCl DESs. The solubility of Cu (I)O and Cu (II)O was observed to be highest in FA-ChCl (greater than 10^6 ppm) which decreases with an increase in alkyl chain length of HBD of DES. CuO dissolution in DES was observed to be independent of polarity of solvent as it was observed to be dissolved more in FA-ChCl than higher polar OA-ChCl. Effect of time and temperature on copper dissolution was evaluated in best-performing FA-ChCl DESs. 100 °C, 18 h or 75 °C, 27 h were optimum condition for maximum copper dissolution. Results of this paper focuses on detailed physiochemical properties of Carboxy A-ChCl DES which can directly be utilized to predict DESs performance in various industrial applications. It covers critical properties of Carboxy A-ChCl DES which can play a significant role in solubility of metals or gases in Carboxy A-ChCl DES. Moreover, mechanism of metal DESs interaction is still lacking in the literature. Theoretical studies could be combined with experimental results to interrogate molecular level interaction of DESs with metals for future studies. Further, Carboxy A-ChCl could be explored as a solvent

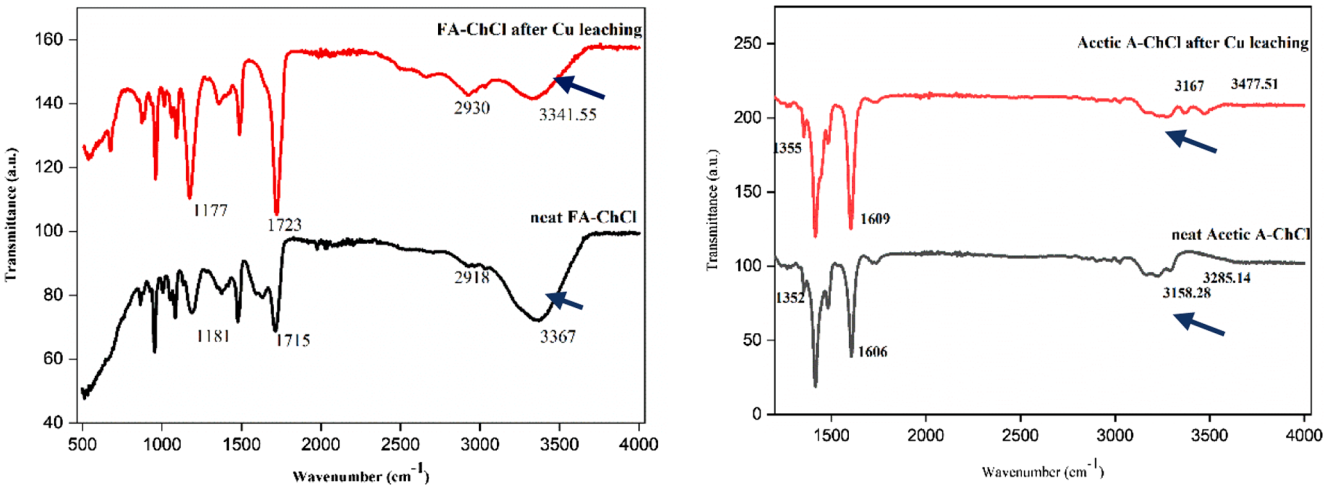


Fig. 8. FTIR spectra of neat FA-ChCl and Cu dissolved FA-ChCl; neat Acetic A-ChCl and Cu dissolved in Acetic A-ChCl.

for other copper sources such as sulphide ores.

CRediT authorship contribution statement

Snigdha Mishra: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Ashish Pandey:** Methodology, Writing – review & editing. **K.K. Pant:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Bhoopesh Mishra:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2023.122142.

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